

Marchal's family: Inclined co-orbitals in the three-body problem

Alexandre Prieur

With Philippe Robutel & Jacques Fejoz

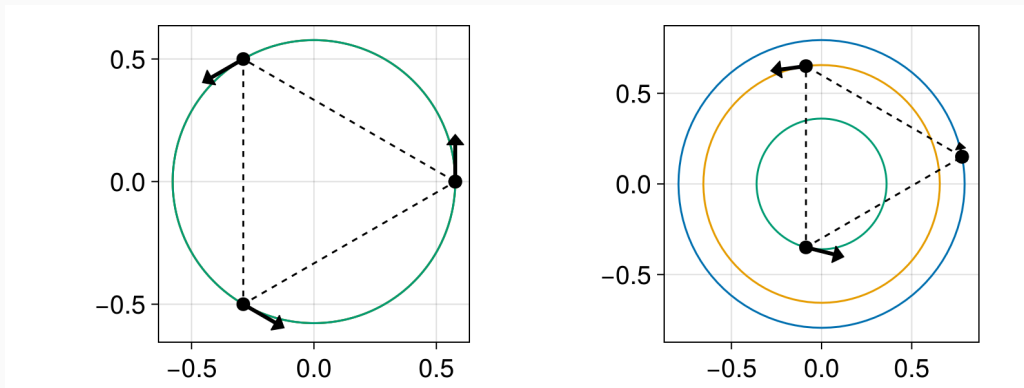
14/09/2026



Introduction



The Lagrange relative equilibrium



Lagrange relative equilibrium of the 3BP

Elliptic if

$$\frac{m_0 * m_1 + m_0 * m_2 + m_1 * m_2}{(m_0 + m_1 + m_2)^2} > \frac{1}{27}$$

$$\begin{cases} (1 - \varepsilon, \varepsilon, 0) \rightarrow \varepsilon_G \approx 0.0385 \\ (1 - 2\varepsilon, \varepsilon, \varepsilon) \rightarrow \varepsilon_G \approx 0.0191 \end{cases}$$

The Lagrange relative equilibrium – Lyapunov families

Lyapunov families emerge from the elliptic directions of Lagrange

3 planar Lyapunov families, one orthogonal to the orbital plane : the **vertical family**

The vertical family – equal masses

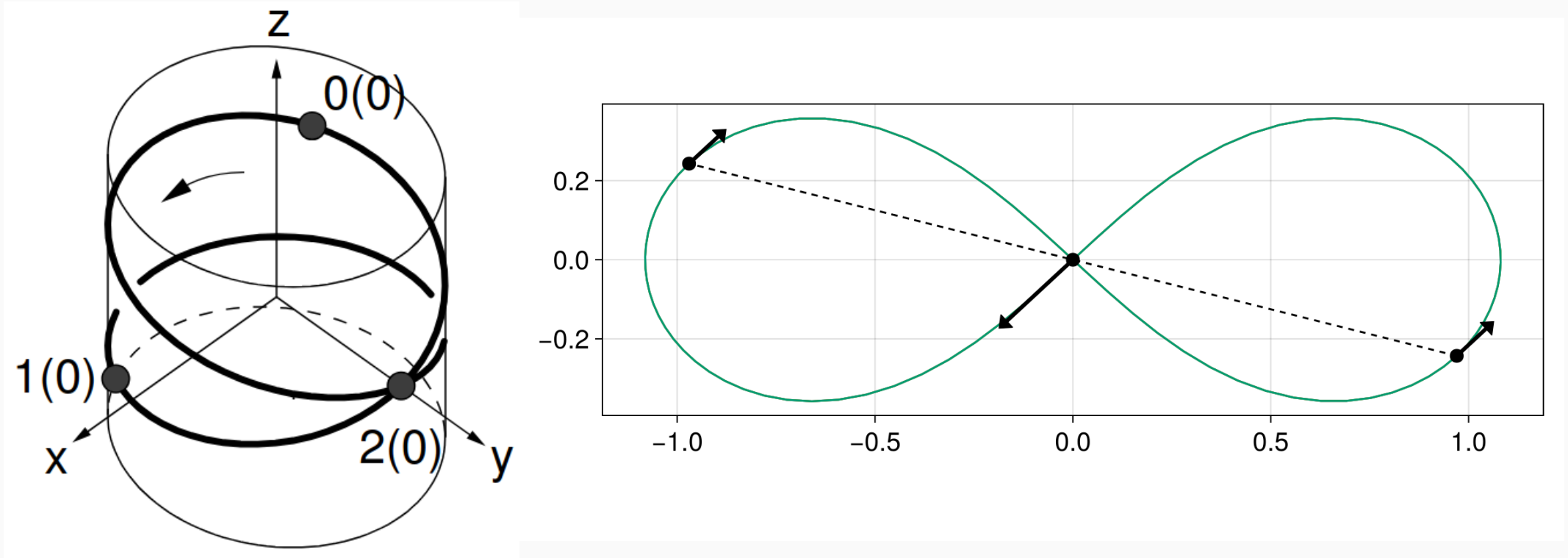
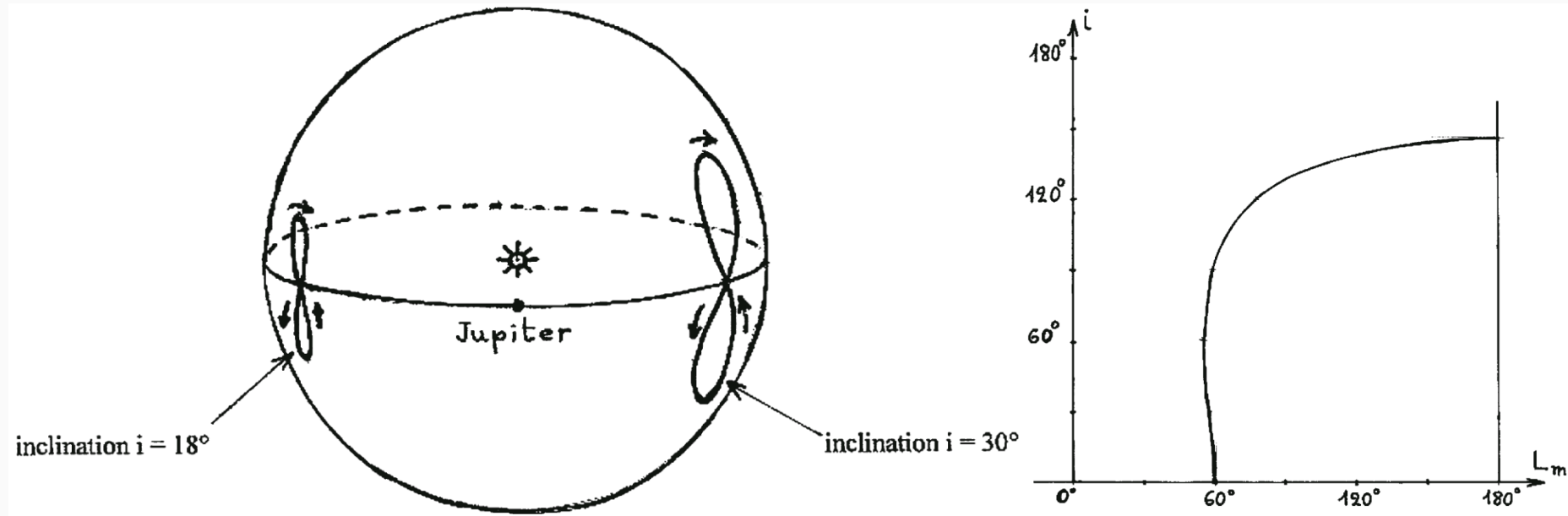


Figure 2: Left: (Chenciner and Fejoz, 2008, fig 3), an orbit of P_{12} in a rotating frame. Right: the Eight.

Link conjectured by Marchal in 2000, proved in (Calleja et al., 2024)

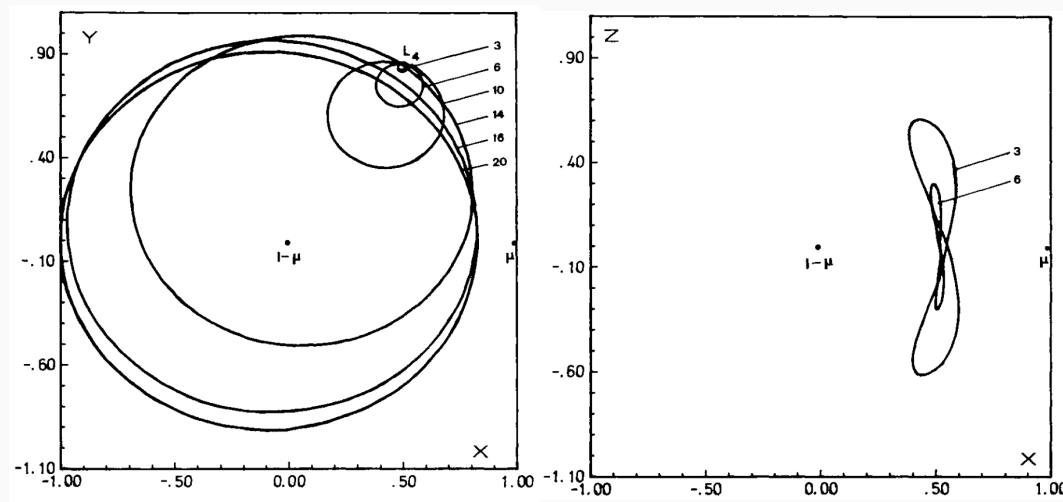
The vertical family – small masses



- (Marchal, 2009) – $\mathcal{VF}_L$ in the average restricted problem
 - Linearly stable in the directions he considers
- (Leleu, 2016) – numerical hints of a stable region in the full planetary problem

A historical note: forgetting previous work

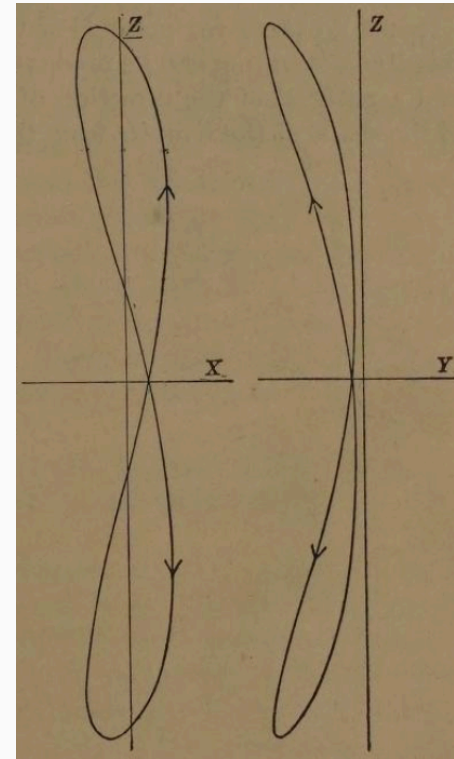
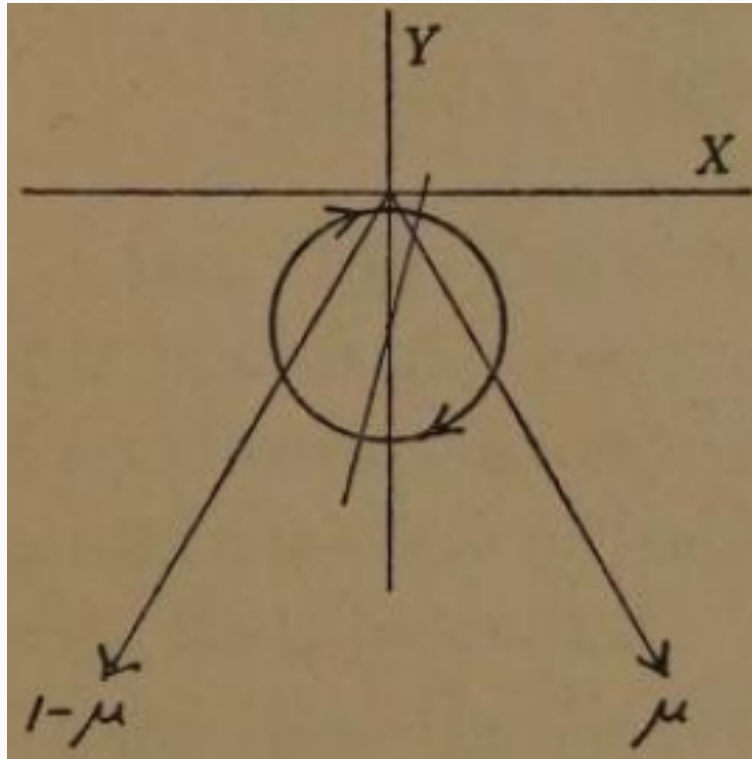
(Marchal, 2009) results are in (Zagouras, 1985)



But (Zagouras, 1985)'s method is described in “Zagouras, C.G.: 1982, Post doctoral dissertation, University of Patras, Greece.”

A historical note: forgetting previous work

... And (Zagouras, 1985) is an extension of (Moulton, 1920), chap IX (by Buck)!



Marchal's generalized conjecture

These periodic orbits belong to a three parameter family of similar periodic orbits of the general three body problem, two parameters being the mass ratios of the three bodies and the third a generalization of the inclination i .
[...] The one parameter family P_{12} of (Marchal, 2001) is another limit case of this three parameter family: the case of three equal masses.

— (Marchal, 2009)

Questions

- Q1: Does the family persist in the non-restricted/non-average problem?
- Q2: What is the stability of the orbits along this family?
- Q3: Does it have an impact on the global stability of the co-orbital region?
- Q4: What is the link between the family and P_{12} ?

$\mathcal{VF}_L$ in the average circular
planetary problem

Frame of reference

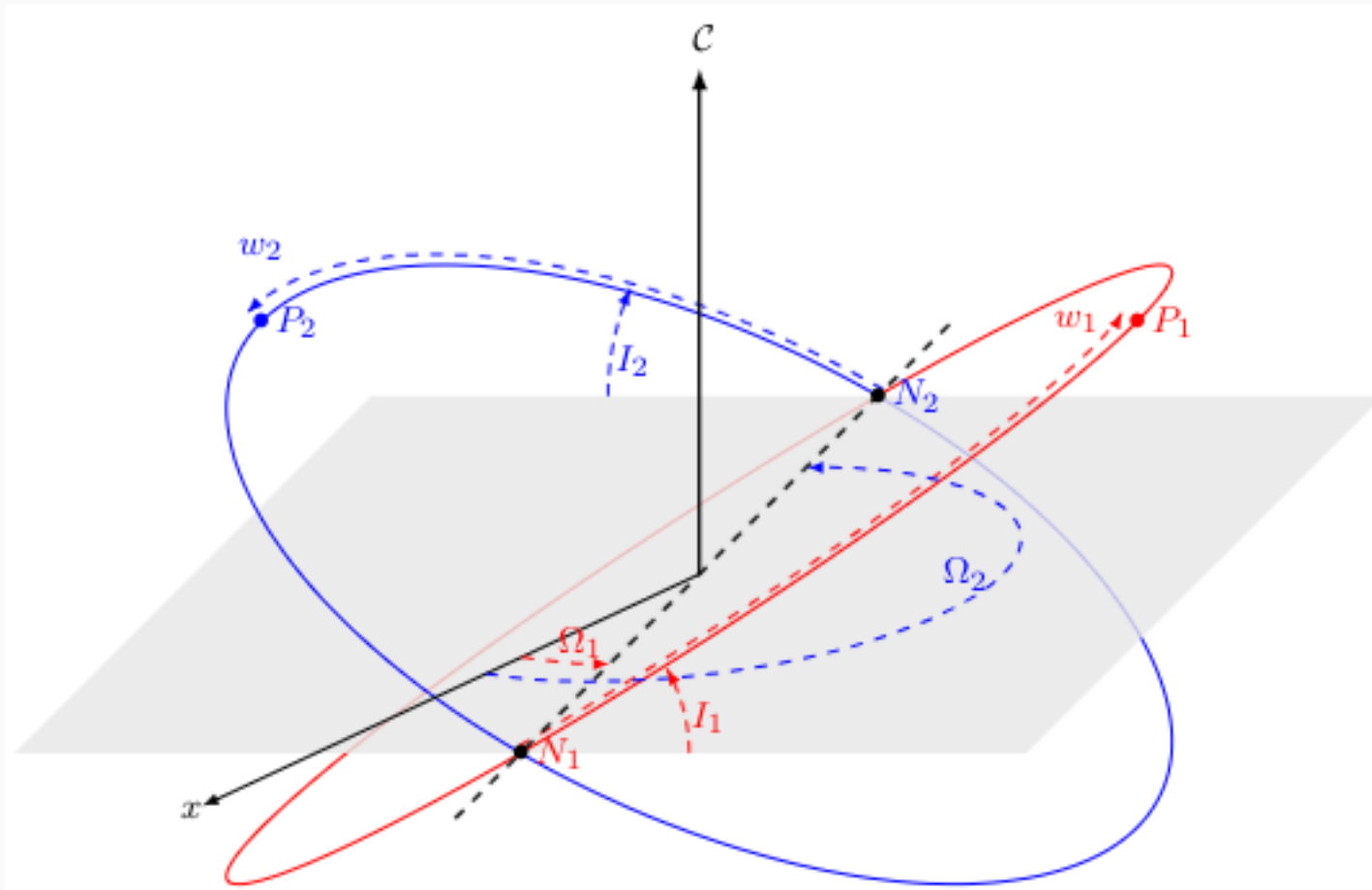


Figure 6: Canonical heliocentric : (Robutel, 2015) heliocentric positions, barycentric momenta; $(Oxy) \perp \mathbf{C}$
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Co-orbital motion in the planetary case

Complex Poincaré-like coordinates

Planetary problem

$$(1 - \varepsilon(m_1 + m_2), \varepsilon m_1, \varepsilon m_2), \varepsilon \ll 1$$

$\sqrt{\varepsilon}$ -neighborhood of co-orbital motions

$$a_1, a_2 = a^* + \mathcal{O}(\varepsilon)$$

Averaging over fast angle ζ_2

The submanifold of eccentricity 0 is invariant under the flow: circular problem

A one-parameter family: $\mathcal{V}\mathcal{F}_L$

$$\bar{F}(Z_1, \zeta_1, C) = \alpha Z_1^2 + \varepsilon \beta F_1(\zeta_1, C)$$

For a fixed C_0 , one-parameter family of equilibrium points satisfying

$$Z_1 = 0, \quad \zeta_1 = \zeta(C_0), \quad \frac{\partial F_1}{\partial \zeta_1}(\zeta(C_0), C_0) = 0$$

F_1 independant of masses! $\Rightarrow$ Study valid for all small masses

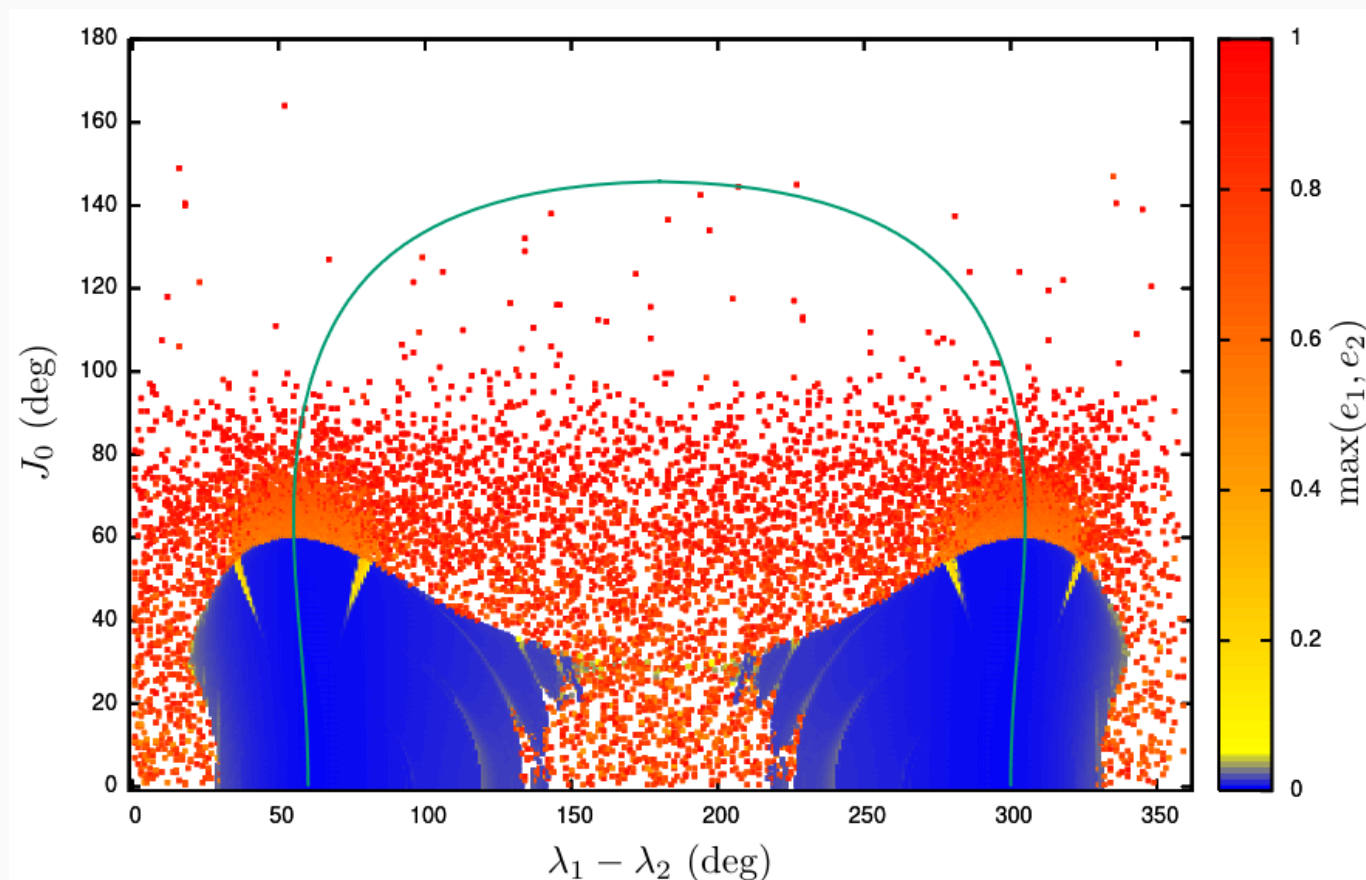
Next ?

- Analytical approximation
- Or numerical search

What can we say?

Analytical: 20^{th} degree expansion in $\sin\left(\frac{J_{\max}}{2}\right)$

Stability: $\mathcal{V}\mathcal{F}_L$ is stable
in the average **circular**
planetary problem



Numerical search in the full problem

A new set of coordinates

- Numerical integration: need coordinates with **explicit** expression of the Hamiltonian/vector field
- The Jacobi reduction should be explicit in these coordinates

Hill coordinates

$$\begin{cases} R_j & r_j \\ G_j & w_j \\ G_j \cos(I_j) & \Omega_j \end{cases}$$

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Hill coordinates

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How to search for periodic orbits?

Looking for zeros of

$$f : \begin{cases} \mathbb{R}^n & \rightarrow \mathbb{R}^n \\ (\mathbf{x}; C) & \mapsto \Phi_T(\mathbf{x}; C) - \mathbf{x} \end{cases}$$

But the zeros aren't locally unique!

Solution: add **sections**

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How to search for periodic orbits?

Looking for zeros of

$$f : \begin{cases} \mathbb{R}^n & \rightarrow \mathbb{R}^n + m \\ (\mathbf{x}; C) & \mapsto \begin{pmatrix} \Phi_T(\mathbf{x}; C) - \mathbf{x} \\ \sigma(\mathbf{x}; C) \end{pmatrix} \end{cases}$$

But the zeros aren't locally unique!

Solution: add **sections**

How to search for periodic orbits?

In our case:

$$\begin{cases} \sigma_1(\mathbf{x}; C) = w_1 + w_2 \\ \sigma_2(\mathbf{x}; C) = J - J_0 \end{cases} \quad \text{Selecting a point on the orbit}$$

Focusing on the case of **two equal masses**: $(1 - 2\varepsilon, \varepsilon, \varepsilon)$

Rootfinding algorithm: multi-dimensional **Newton-Raphson** method

Numerical integration with `taylor` software (**Jorba and Zou, 2005**) via

`TaylorInterface.jl` (in **julia** (**Bezanson et al., 2017**))

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Small masses: $\varepsilon = 10^{-3}$

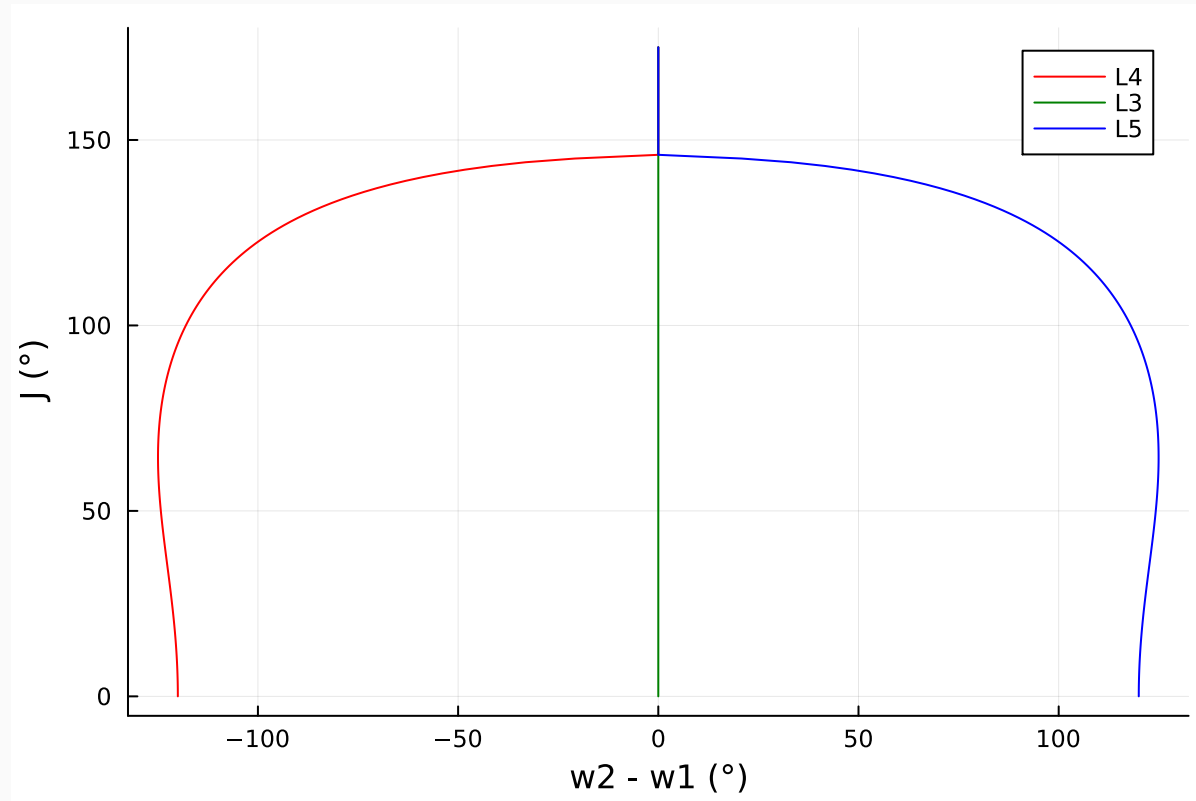
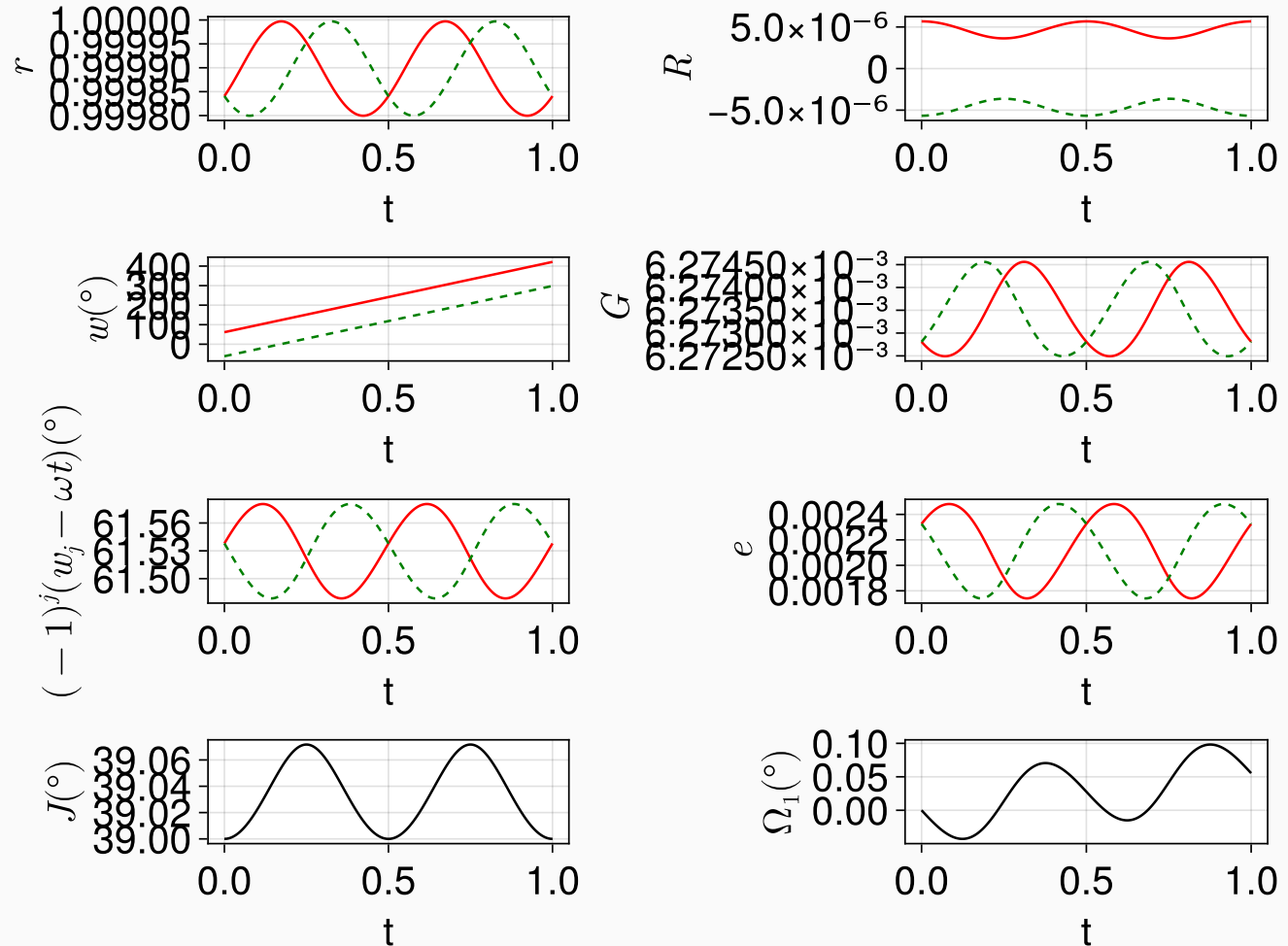


Figure 8: $\mathcal{VF}_L$ from L_4 and L_5 , $\mathcal{VF}_E$ from L_3

Coordinates along trajectory at $\varepsilon = 10^{-3}$

Coordinates at $J = 39^\circ$



Trajectory examples

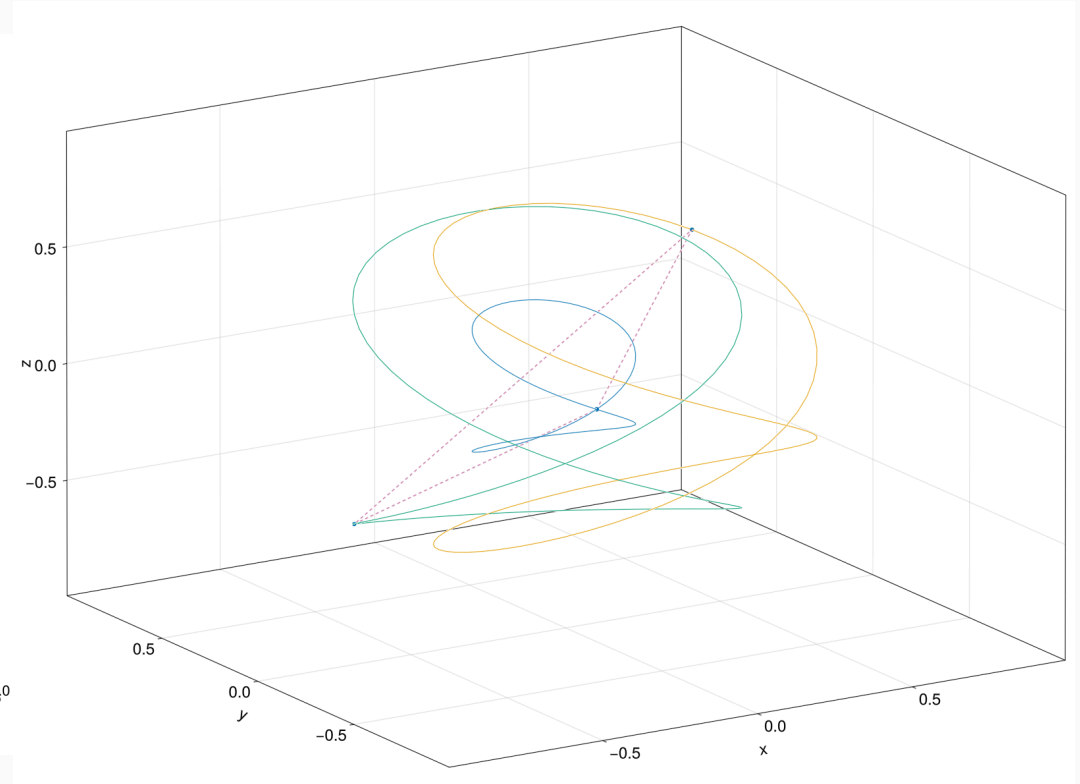
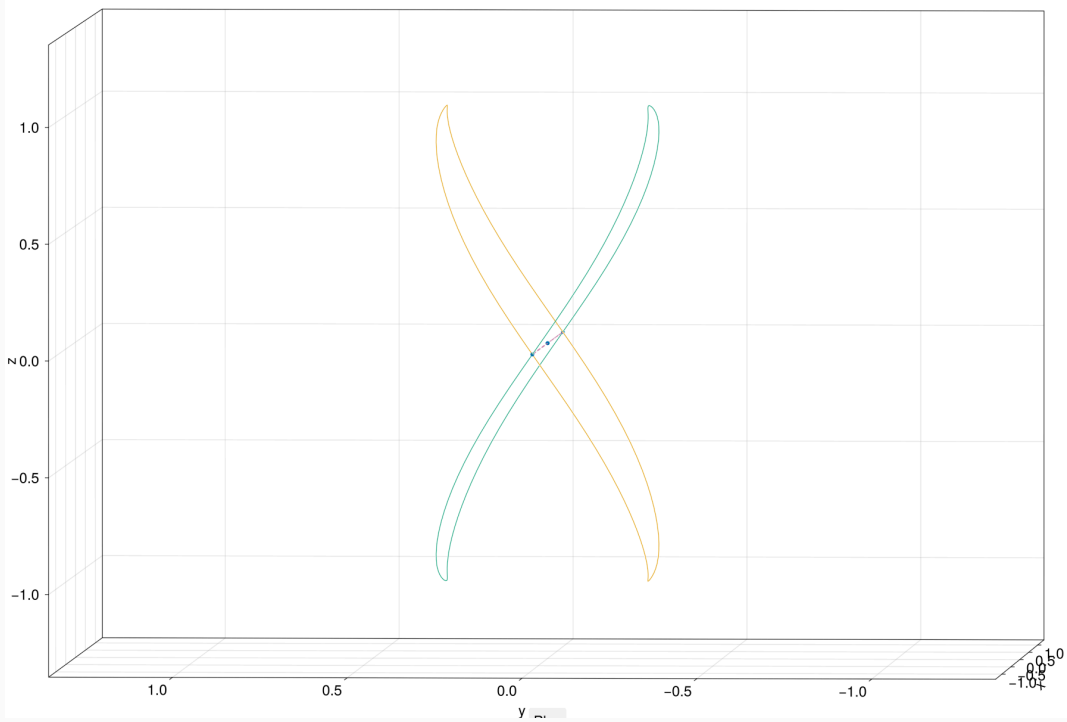
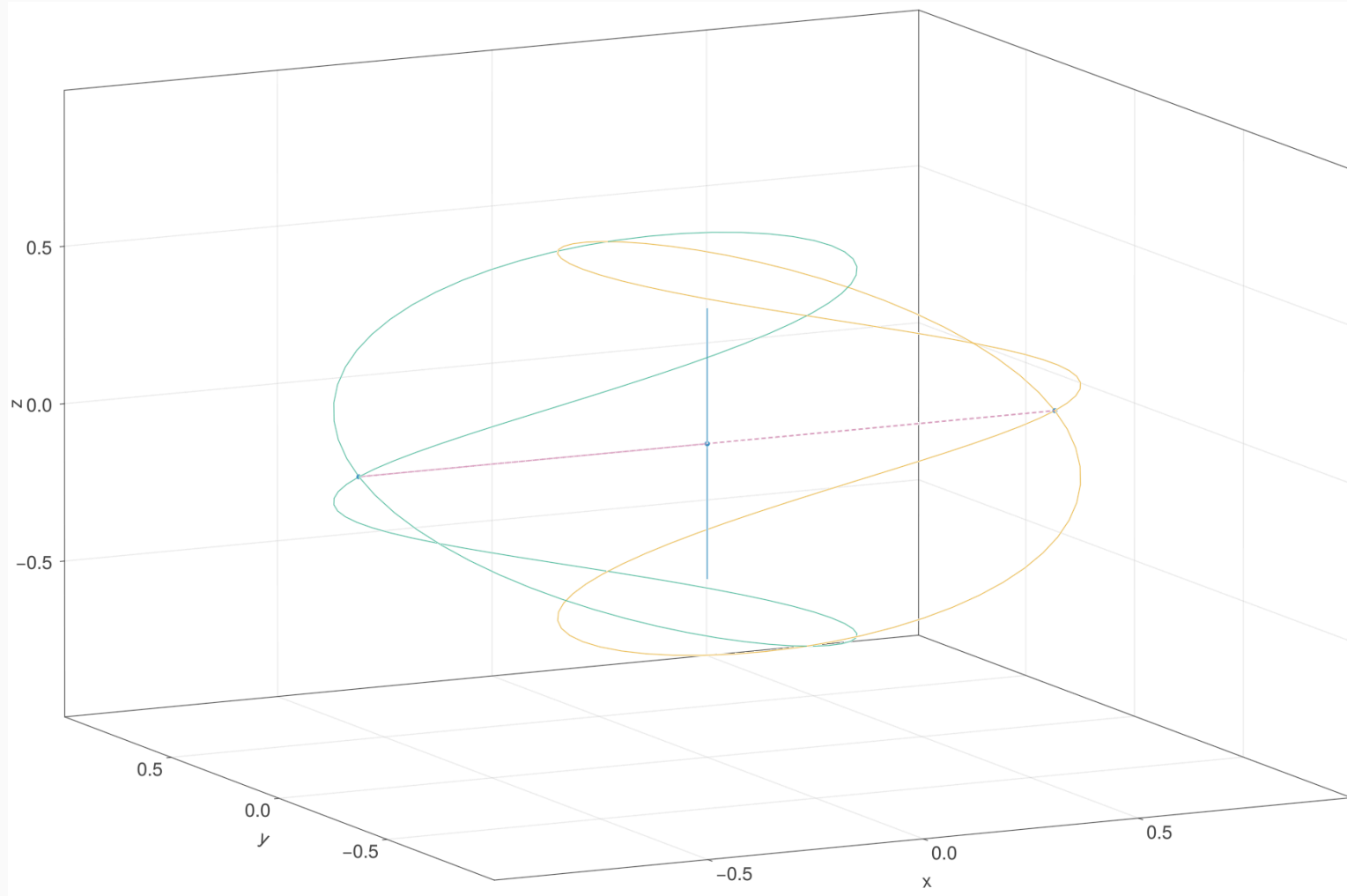


Figure 10: $\mathcal{VF}_L$ orbits in the draconic (left) and uniformly rotating (right) plane, for $\varepsilon = 0.21, J = 106^\circ$.

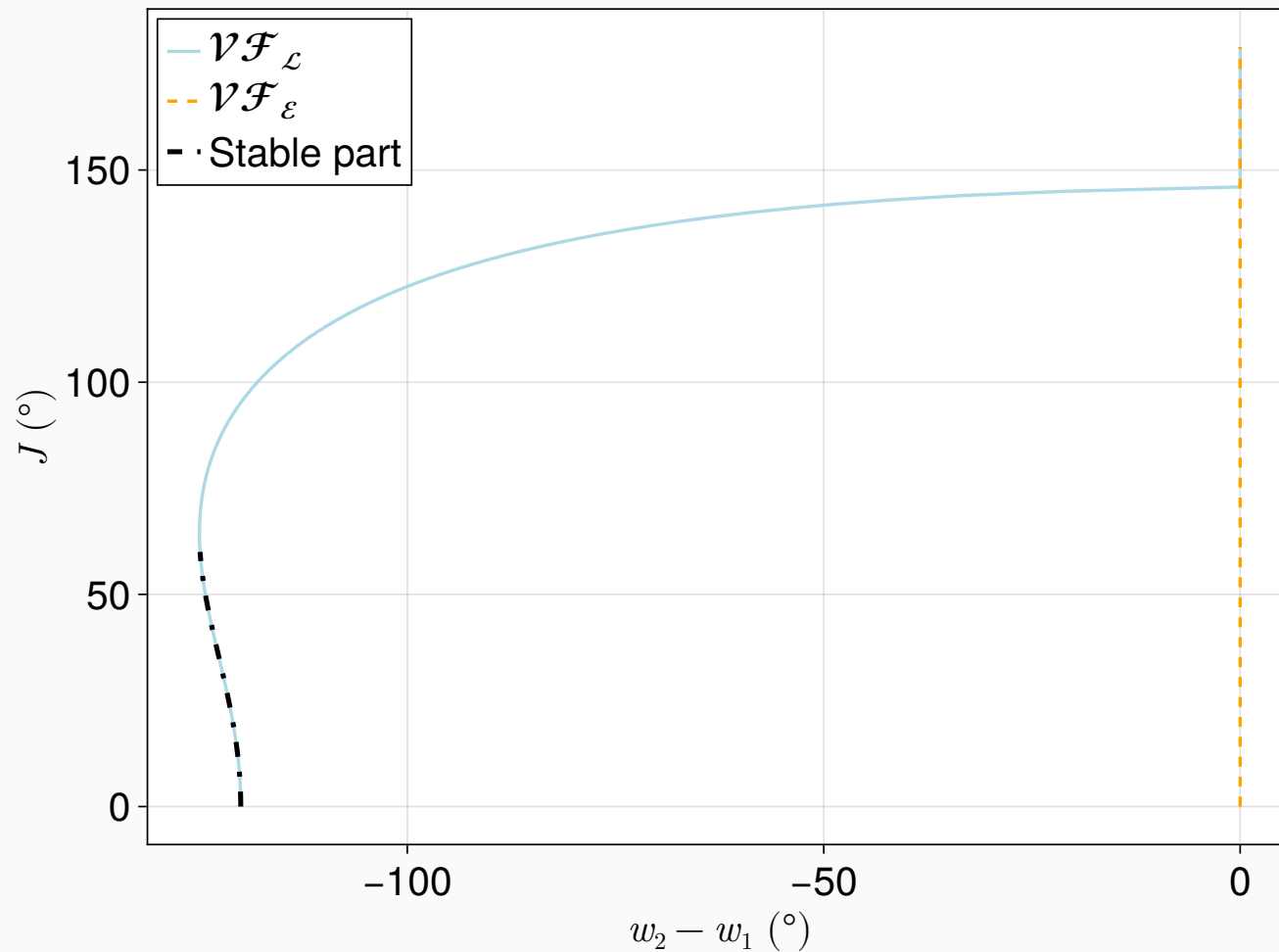
Trajectory example – $\mathcal{VF}_E$



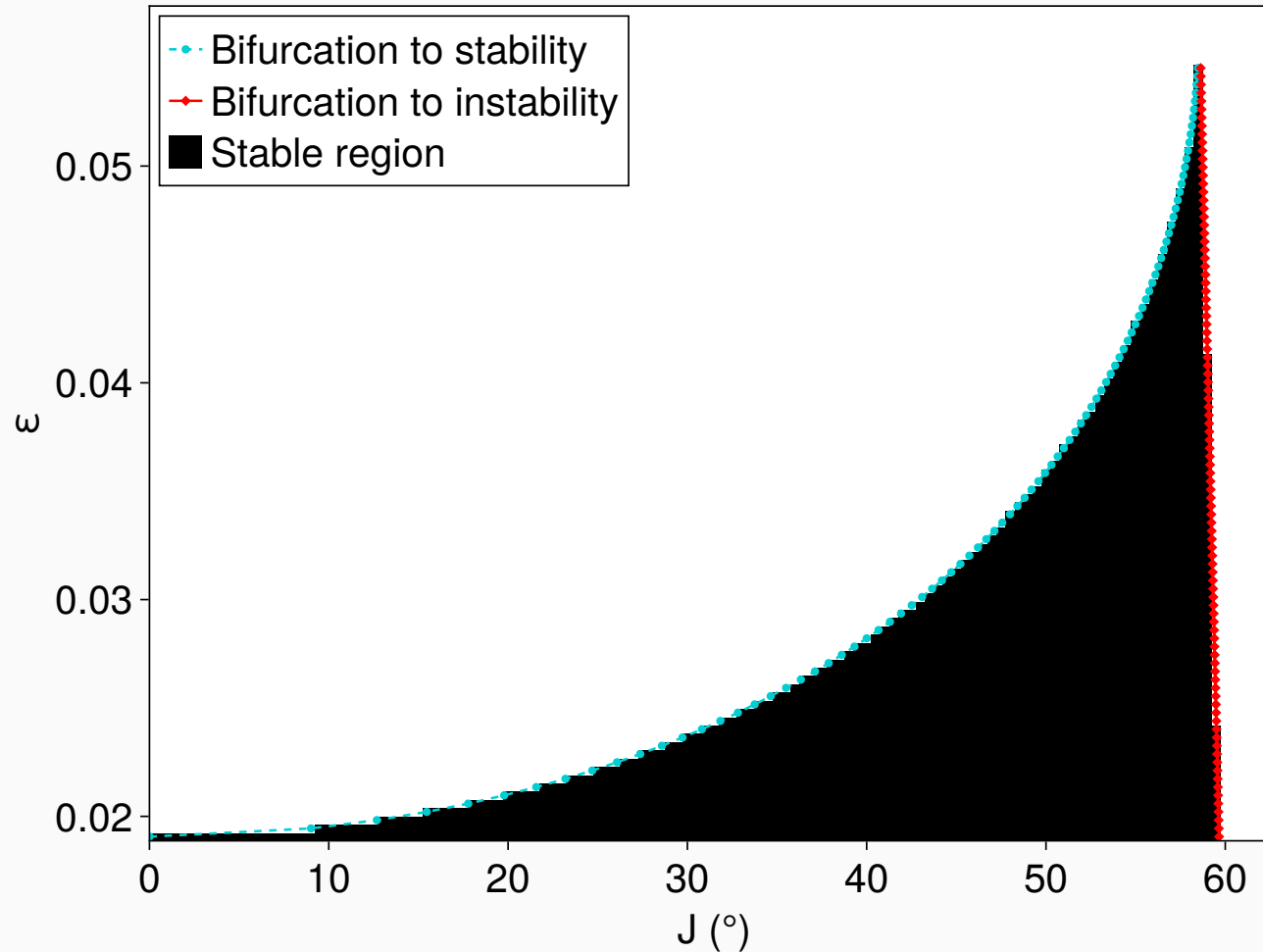
Stability



Stability at $\varepsilon = 10^{-3}$

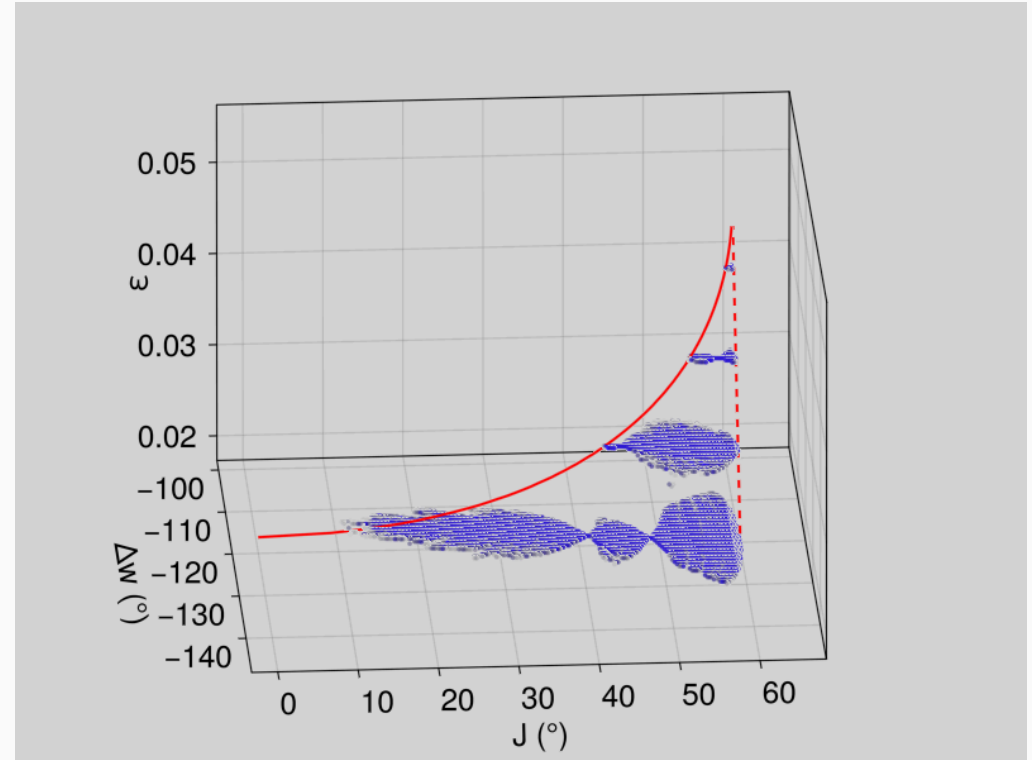
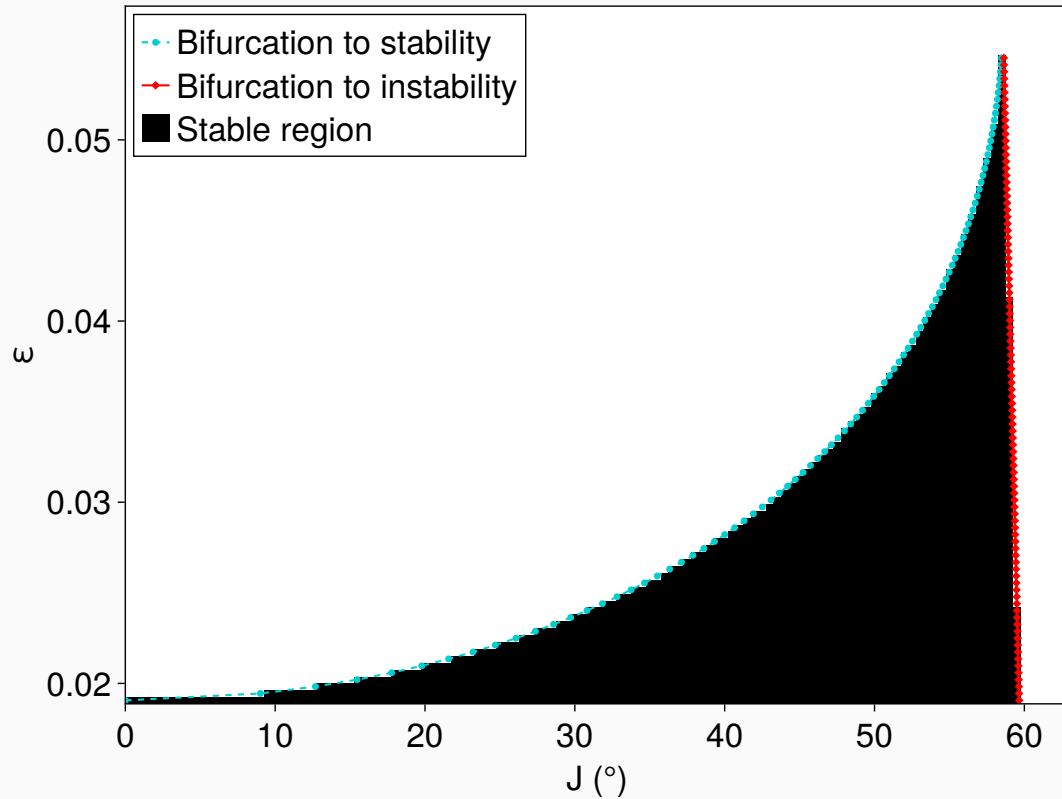


Increasing masses: above Gascheau's value

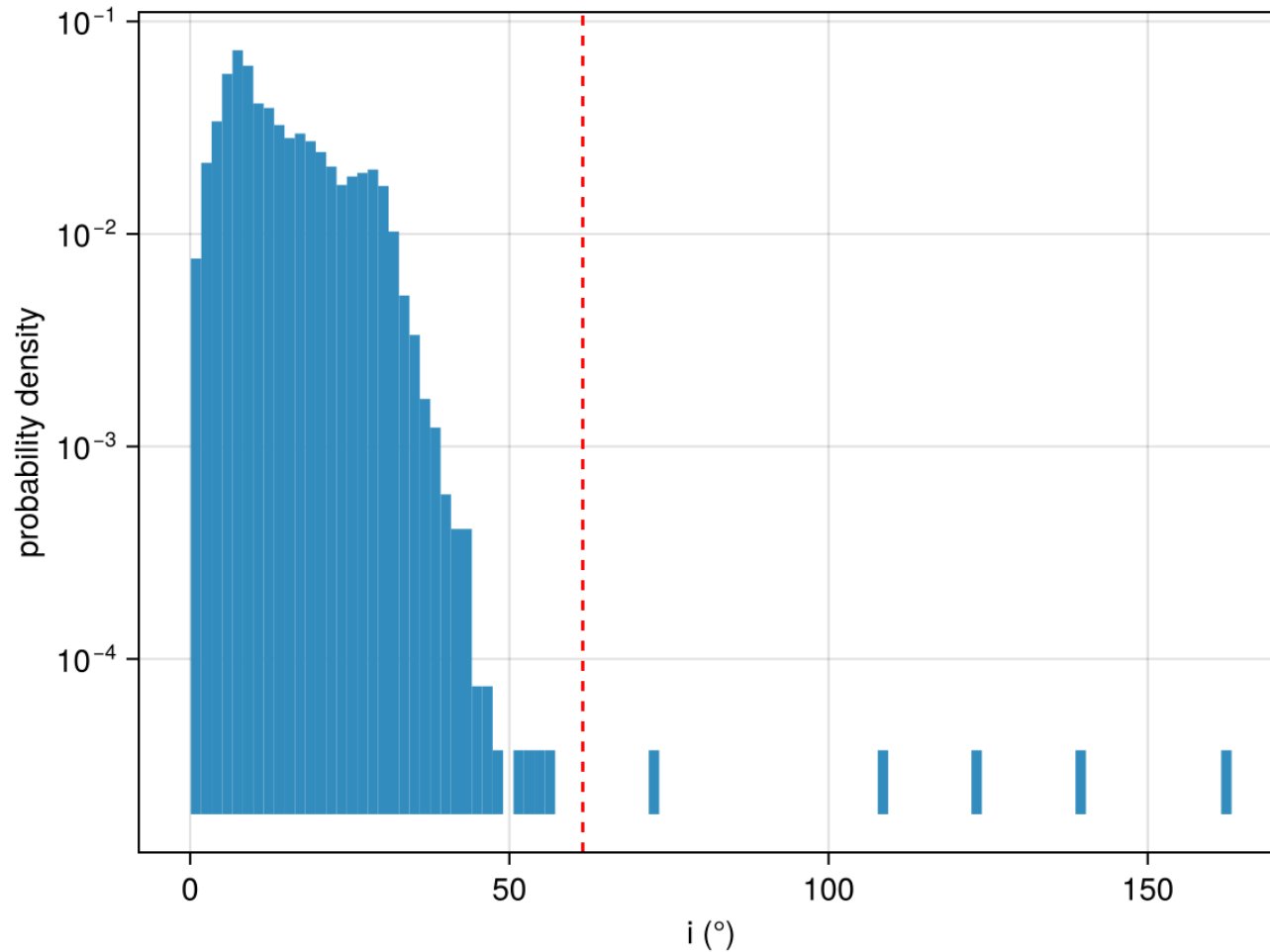


Similar to (Roberts,
2002)

Above Gascheau's value (long-term stability)



Astronomical consequences?



Towards equal masses

Symmetries

Half-time symmetry

$$\forall \mathbf{x} \in \mathcal{VF}_L, \Phi_{\frac{T}{2}}(\mathbf{x}; C) = -\mathbf{x}$$

$$\Rightarrow f : (\mathbf{x}; C) \mapsto \begin{pmatrix} \Phi_{\frac{T}{2}}(\mathbf{x}; C) + \mathbf{x} \\ \sigma_1(\mathbf{x}; C) \\ \sigma_2(\mathbf{x}; C) \end{pmatrix}$$

Cuts integration time in half

Time-reversal symmetry

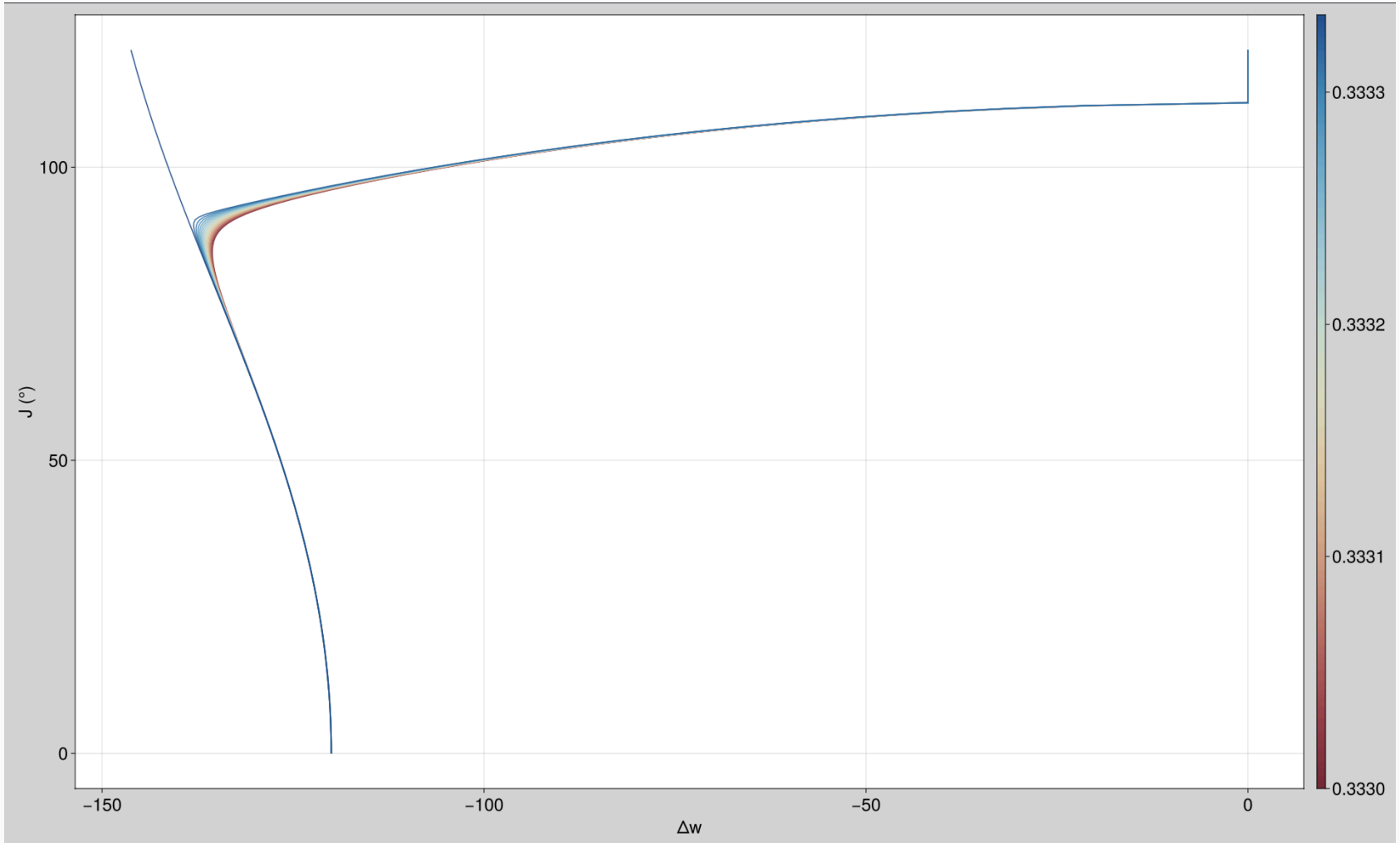
$$\forall \mathbf{x} \in \mathcal{VF}_L, t \in \mathbb{R},$$

$$\Phi_{-t}(\mathbf{x}; C) = R_y(\pi) p_{1 \leftrightarrow 2} \Phi_t(\mathbf{x}; C)$$

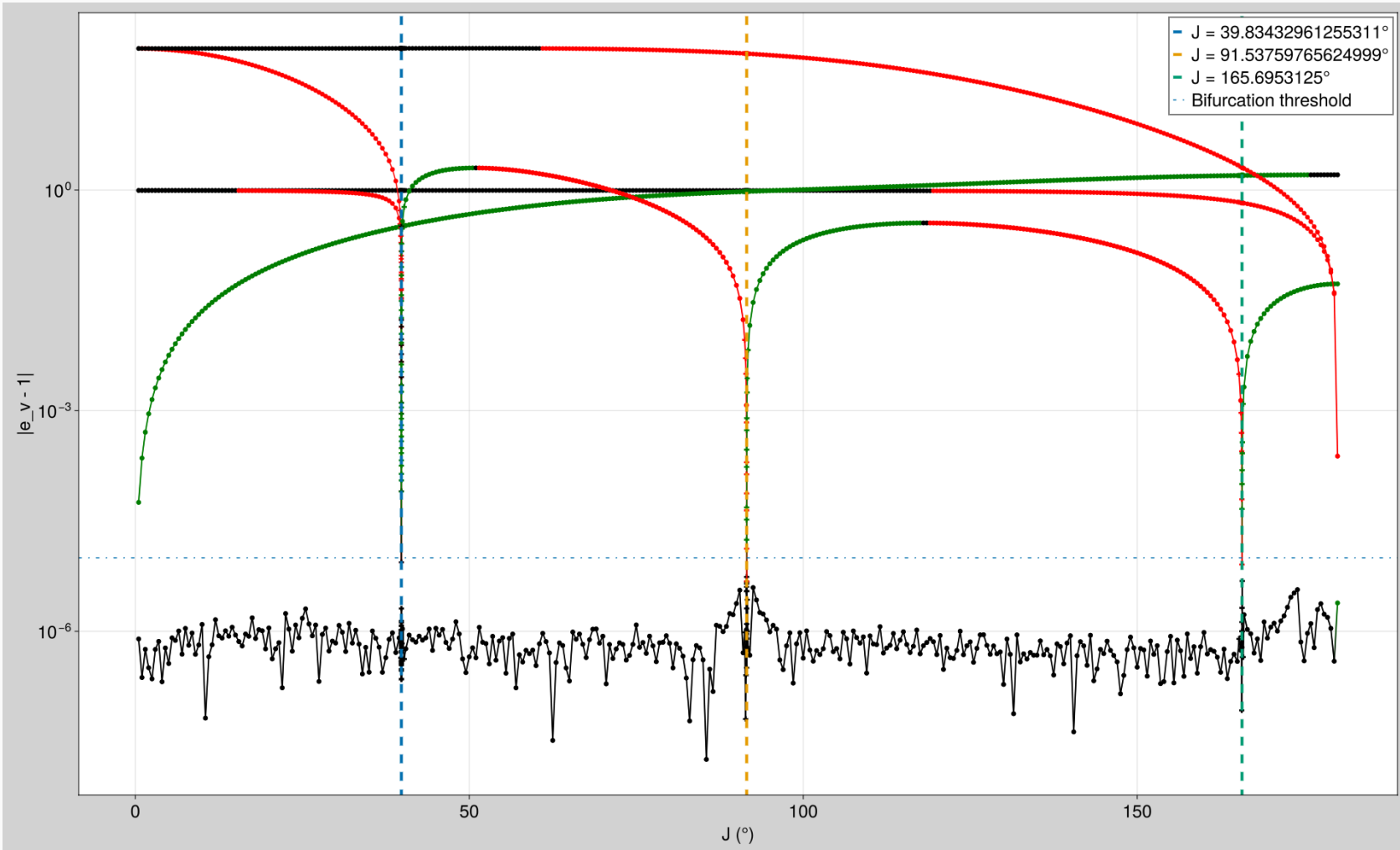
Allows to add sections

$$\sigma_3(\mathbf{x}; C) = \begin{pmatrix} r_1 - r_2 \\ R_1 + R_2 \\ G_1 - G_2 \end{pmatrix}$$

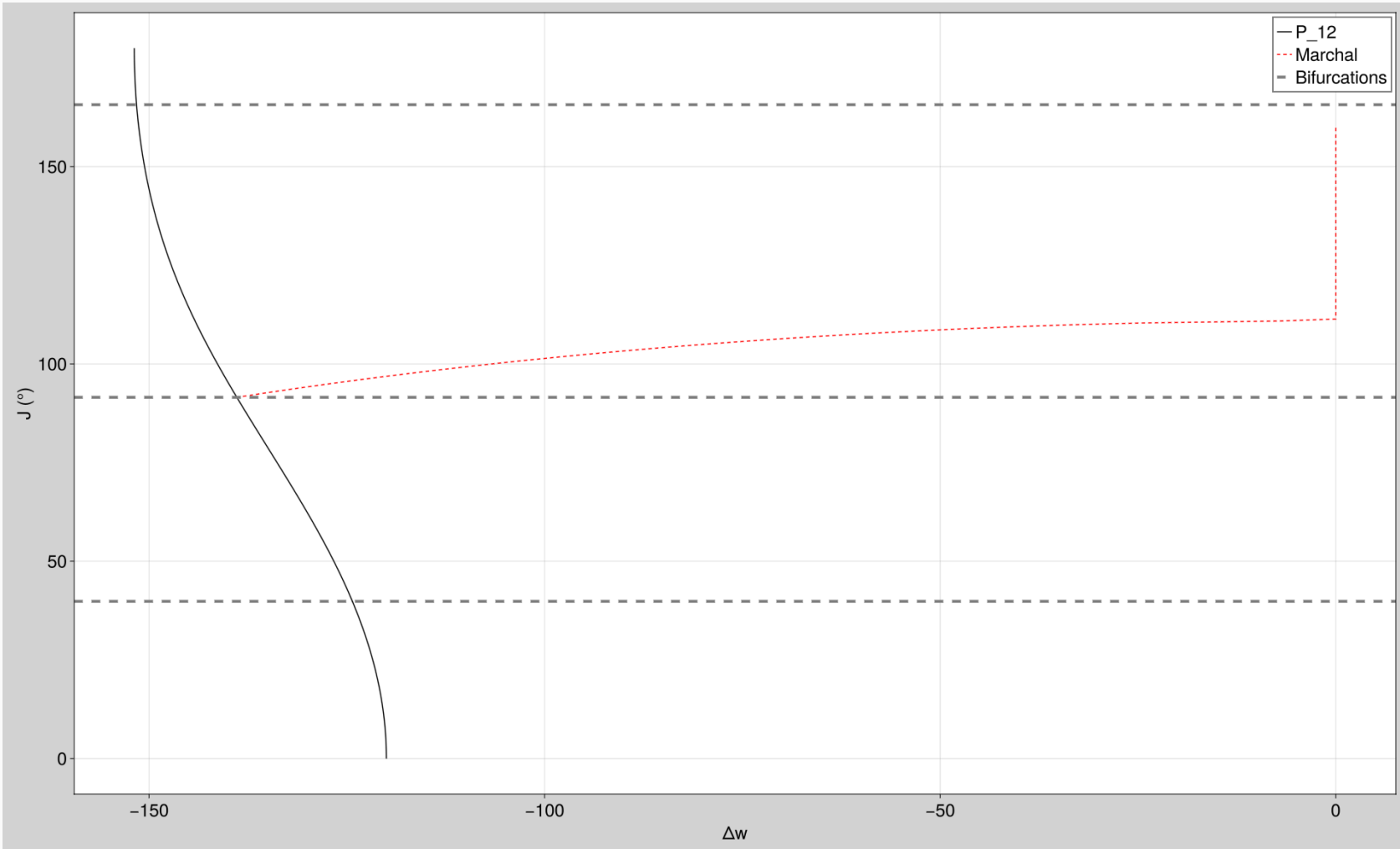
Approaching equal masses



Bifurcations at equal masses



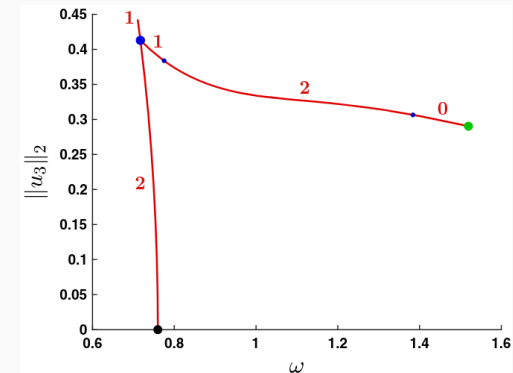
$\mathcal{V}\mathcal{F}_L$ and P_{12} : Marchal's family



The other bifurcations



- Third bifurcation unexplored for now
- First bifurcation to a “lazy Eight”
 - Cf. (Calleja et al., 2021), and Hénot, Fejoz & Chenciner



Bifurcations at non-equal masses

These bifurcations also exist at low masses!

P_{12} confirmed at almost-equal masses ($\frac{1}{3} - \varepsilon \approx 10^{-5}$)

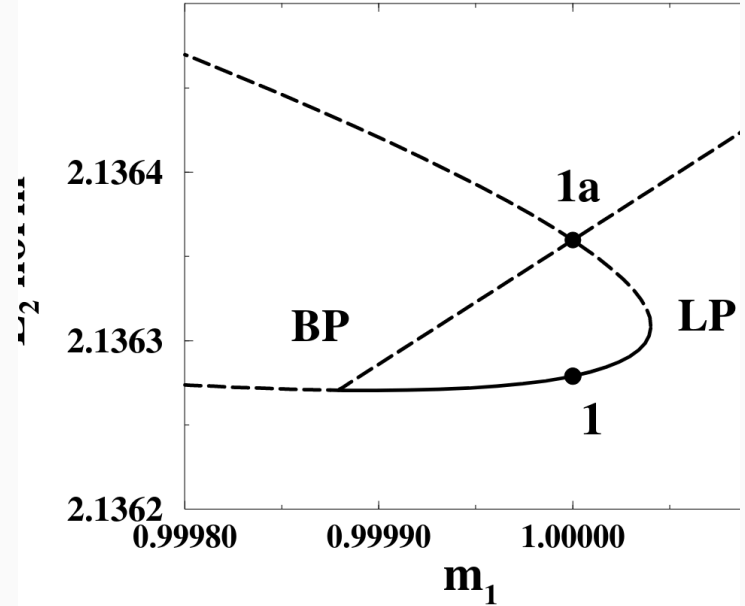
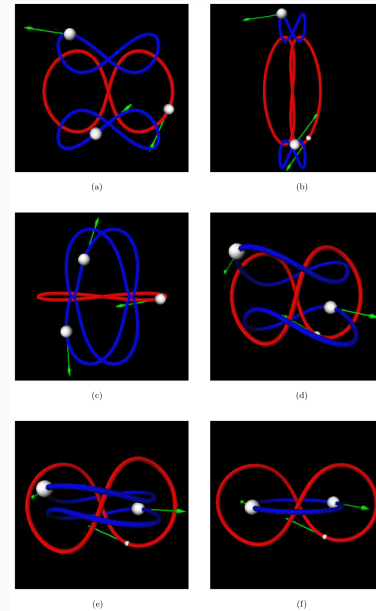
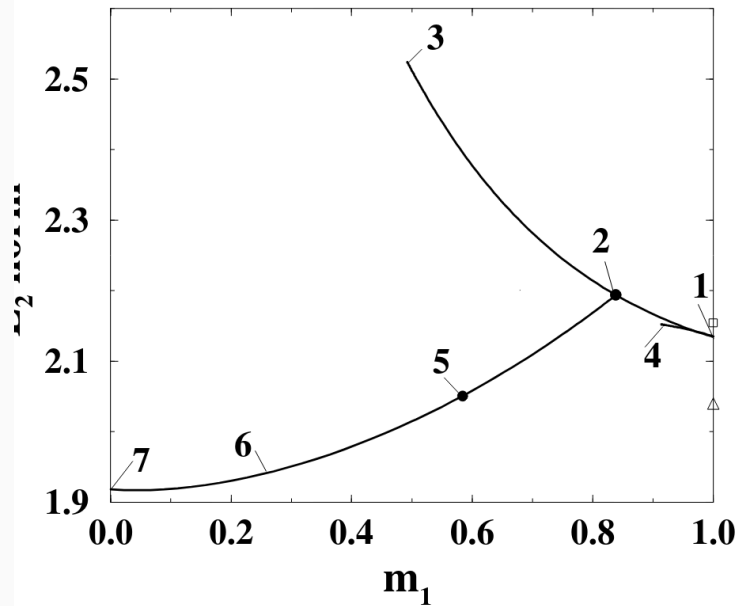


Figure 22: (Doedel et al., 2003)

Conclusion

Further work: bifurcations

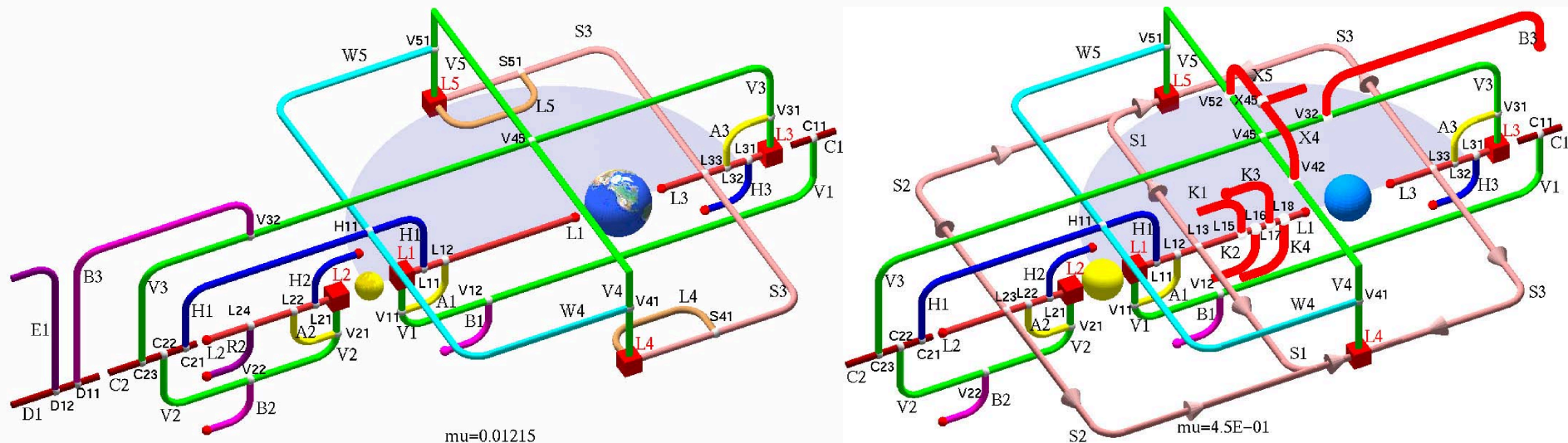


Figure 23: (Doedel et al., 2007, fig 3 and 51).

- Detailed study and following of bifurcations along $\mathcal{VF}_L$ and P_{12}
 - Continue P_{12} at non-equal masses
- Apply `TaylorInterface.jl` and our numerical toolbox to more problems

Conclusion

We showed...

- Persistence of $\mathcal{VF}_L$ in the full problem
- Numerical validation of Marchal's generalized conjecture, up to bifurcations

And we developed an open-source numerical toolbox for investigating recurring dynamics

Much remains to be done!

In memory of Àngel and Christian

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